

Laws of motion

When driving on curved roads, it's important to understand the forces affecting your vehicle and adjust your driving to stay in control.

- **Roll and yaw.** Roll affects how much your vehicle tilts during a turn, yaw affects side-to-side movement, and both are important for maintaining control. Adjusting your speed and steering can help balance your roll and yaw while driving through curves.
- **Momentum.** When you're driving in a straight line, your vehicle has momentum in that direction. However, when you enter a curve, you need to change the direction and momentum of your vehicle.
- **Inertia.** In a curve, your vehicle will want to pull a little to keep you moving in a straight line. If you suddenly turn the steering wheel, inertia will try to keep the car moving straight. This is why you must carefully steer to smoothly make the turn.
- **Kinetic energy.** Higher speeds create more energy, requiring more time and space to stop or change direction. Slowing down for curves lowers your vehicle's kinetic energy and helps you make the turn without losing traction or control.
- **Friction.** Friction is the force between the tires and the road. It allows your vehicle to grip the surface. The more friction there is, the better your tires can grip the road. This helps you slow down, speed up, or turn safely. Slippery roads due to rain, snow, or ice can significantly reduce friction in curves. Slow down even more and avoid sudden movements in these conditions.